

MURAFIQ — مرافق — دراسة اقتصاديات وتكاليف الذكاء الاصطناعي

Document: AI Unit Economics & Plan Budget Matrix  
Date: October 2026 | FX Standard: 1 USD = 50.00 EGP

1. OPERATIONAL MODEL RATES & UNIT COSTS (AT 50.00 EGP / USD)

| SERVICE COMPONENT                   | AI MODEL / PROVIDER                          | OPERATION TYPE                                          | UNIT RATE (USD) | UNIT RATE (EGP)           | BILLING TRIGGER                                |
|-------------------------------------|----------------------------------------------|---------------------------------------------------------|-----------------|---------------------------|------------------------------------------------|
| AI Stylist Chat Message             | Google Gemini 3.1 Flash Lite                 | Intent extraction + Outfit composition + Render         | \$0.0020 USD    | 0.100 EGP (10 Piastres)   | Per completed styling turn                     |
| Wardrobe Image Classification       | Google Gemini 3.1 Flash Lite (Vision)        | Extract 14 garment attributes + category + colors + fit | \$0.0007 USD    | 0.035 EGP (3.5 Piastres)  | Once at upload (persisted forever in DB)       |
| External Outfit Search (Serper.dev) | Serper Google Shopping Egypt (Active)        | Live Egyptian retailer search (\$50 / 50k queries pool) | \$0.0010 USD    | 0.050 EGP (5.0 Piastres)  | Per live search query (3 queries per gap call) |
| External Outfit Search (SerpApi)    | SerpApi Google Shopping (Benchmark)          | Live Egyptian retailer search (\$50 / 5k queries pool)  | \$0.0100 USD    | 0.500 EGP (50.0 Piastres) | Per live web query (Monthly SaaS)              |
| Virtual Try-On (TryOn 15F)          | Nano Banana Models (Gemini Flash Image Lite) | Shape Model + garments composite image generation       | \$0.0336 USD    | 1.680 EGP (168 Piastres)  | Per generated try-on image (Paid tiers only)   |

2. MASTER SUBSCRIPTION PLANS & COST MATRIX (ALL COMPONENTS ITEMIZED)

| PLAN & PRICE                           | AI CHAT MESSAGES                                | WARDROBE CLASSIFICATION             | PRODUCT SEARCHES (SERPER)              | NANO BANANA (TRY-ON)                   | TOTAL AI COST (USD / EGP) | NET PROFIT (EGP) | MARGIN      |
|----------------------------------------|-------------------------------------------------|-------------------------------------|----------------------------------------|----------------------------------------|---------------------------|------------------|-------------|
| Client Free<br>0 EGP / \$0.00          | 10 total / account<br>1.00 EGP (\$0.020)        | 7 items max<br>0.25 EGP (\$0.005)   | 0 searches<br>0.00 EGP                 | 0 try-ons (Disabled)<br>0.00 EGP       | 1.25 EGP<br>\$0.025 USD   | -1.25 EGP        | CAC (Trial) |
| Client Basic<br>50 EGP / \$1.00        | 5 / day (~150/mo)<br>15.00 EGP (\$0.300)        | 25 items max<br>0.88 EGP (\$0.0175) | 15 searches / mo<br>2.25 EGP (\$0.045) | 4 try-ons / mo<br>6.72 EGP (\$0.134)   | 24.85 EGP<br>\$0.497 USD  | +25.15 EGP       | +50.3%      |
| Client Mid<br>150 EGP / \$3.00         | 35 / day (~700 real)<br>70.00 EGP (\$1.400)     | 45 items max<br>1.58 EGP (\$0.0315) | 30 searches / mo<br>4.50 EGP (\$0.090) | 6 try-ons / mo<br>10.08 EGP (\$0.202)  | 86.16 EGP<br>\$1.723 USD  | +63.84 EGP       | +42.6%      |
| Client Pro<br>250 EGP / \$5.00         | 80 / day (~1,400 real)<br>140.00 EGP (\$2.800)  | 100 items max<br>3.50 EGP (\$0.070) | 45 searches / mo<br>6.75 EGP (\$0.135) | 8 try-ons / mo<br>13.44 EGP (\$0.269)  | 163.69 EGP<br>\$3.274 USD | +86.31 EGP       | +34.5%      |
| Client Enterprise<br>500 EGP / \$10.00 | 150 / day (~2,500 real)<br>250.00 EGP (\$5.000) | 200 items max<br>7.00 EGP (\$0.140) | 60 searches / mo<br>9.00 EGP (\$0.180) | 10 try-ons / mo<br>16.80 EGP (\$0.336) | 282.80 EGP<br>\$5.656 USD | +217.20 EGP      | +43.4%      |

**1. Wardrobe Classification**

Garments are classified asynchronously via BullMQ. Because it runs **strictly once per garment** and caches normalized attributes in MongoDB, classifying 25 items costs only **0.88 EGP**, leaving chat reasoning free forever.

**2. Unit Economics on Basic**

Capping the Basic tier (50 EGP) at 5 msgs/day, 15 searches, and 4 try-ons with Serper delivers a solid **+50.3% profit margin (+25.15 EGP)**, protecting company cash flow.

**3. Zero Try-On Leak on Free**

Disabling Virtual Try-On on the Free tier sets customer acquisition cost (CAC) strictly to **1.25 EGP (\$0.025 USD)** for 10 lifetime chat messages and 7 wardrobe slots, eliminating trial GPU waste.

**4. Serper 10x Advantage**

Using Serper.dev (\$0.001/query) instead of SerpApi (\$0.010) or Google Grounding (\$0.0153) keeps search costs under **2.25 EGP / month** on Basic, saving thousands of pounds monthly at scale.

